THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 3: **The World of Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which statement is correct about carbon dioxide? 1

(a) it does not support burning **(b)** it supports burning
(c) it burns **(d)** it is a base

2. An object is dropped on the floor and gives a ringing sound. It is made of: 1

(a) sulphur **(b)** wood **(c)** metal **(d)** chalk

3. Which one produces a basic solution when dissolved in water? 1

(a) nitrogen dioxide **(b)** sulphur dioxide **(c)** calcium oxide **(d)** carbon dioxide

4. A basic oxide reacts with acid to form: 1

(a) gas and metal **(b)** salt and hydrogen
(c) salt and water **(d)** acid and water

5. A shiny black solid conducts electricity, is soft and is used as a pencil lead. It is: 1

(a) graphite **(b)** coal tar **(c)** iron **(d)** sulphur

6. Which oxide gives an acidic solution with water? 1

(a) sodium oxide **(b)** magnesium oxide **(c)** sulphur dioxide **(d)** calcium oxide

7. Which of the following is both a metal and a poor conductor of heat? 1

(a) lead **(b)** silver **(c)** copper **(d)** aluminium

8. Why does the green coating of copper form while aluminium forms a thin white layer? 1

(a) coating is paint **(b)** different reactions with air
(c) aluminium is liquid **(d)** copper is a gas

9. Rust formed on iron in moist air is mainly: 1

(a) an oxide of iron **(b)** iron chloride **(c)** iron sulphide **(d)** iron carbonate

10. Which statement best explains why the Iron Pillar has not rusted much? 1

(a) protective layer and dry climate **(b)** it is hollow
(c) it is made of gold **(d)** it is painted

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why the copper vessel turns black when heated in air. 2

12. How can you show that the oxide of magnesium is basic using turmeric? 2

13. Why are iron nails coated with zinc or paint? 2

14. Why is mercury used in thermometers and barometers? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why are cooking vessels made of steel and not of iron? 3

16. Why do overhead metal wires sag more in summer than in winter? 3

17. How is carbon dioxide used in fire extinguishers and why? 3

OR

Explain how you could separate a mixture of iron filings and sulphur powder using a magnet. 3

18. Why is a zinc coating given to iron buckets? 3

OR

What are alloys? Give two examples with components. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the effects of corrosion on metals and structures and how to reduce the loss. 5

OR

A student heats iron filings and sulphur powder together. Describe the changes and how to prove a new substance has formed. 5

20. Explain why metals conduct heat and electricity but non-metals generally do not, using only observations. 5

OR

Discuss the recycling of metals and its benefits for environment and economy. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND